

The KdV Equation: History, Physics, and Scaling

Historical Origins

- ▶ 1834: John Scott Russell observes a “wave of translation” in a canal.
- ▶ Key features:
 - ▶ Stable shape over long distances
 - ▶ Speed depends on amplitude
- ▶ 1895: Korteweg and de Vries derive an equation modeling shallow water waves.
- ▶ Their goal: balance of
 - ▶ Nonlinearity (steepening)
 - ▶ Dispersion (wave spreading)

Physical Setup

We consider:

- ▶ Inviscid, incompressible fluid
- ▶ Irrotational flow
- ▶ Free surface at $y = \eta(x, t)$
- ▶ Flat bottom at $y = -h$

Velocity potential:

$$\phi(x, y, t), \quad \nabla^2 \phi = 0 \quad \text{for} \quad -h < y < \eta(x, t)$$

Boundary Conditions

Bottom (no penetration):

$$\phi_y = 0 \quad \text{at } y = -h$$

Kinematic condition:

$$\eta_t + \phi_x \eta_x = \phi_y \quad \text{at } y = \eta(x, t)$$

Dynamic condition (Bernoulli):

$$\phi_t + \frac{1}{2} |\nabla \phi|^2 + g\eta = 0 \quad \text{at } y = \eta(x, t)$$

Key Physical Parameters

Introduce:

- ▶ Typical amplitude: a
- ▶ Typical wavelength: λ
- ▶ Depth: h

Dimensionless parameters:

$$\epsilon = \frac{a}{h}, \quad \delta = \frac{h}{\lambda}$$

Shallow water regime:

$$\delta \ll 1, \quad \epsilon \ll 1$$

KdV scaling assumption:

$$\epsilon \sim \delta^2$$

Physical Meaning of Scaling

- ▶ $\epsilon \ll 1$: weak nonlinearity
- ▶ $\delta \ll 1$: long waves relative to depth
- ▶ $\epsilon \sim \delta^2$:
 - ▶ Nonlinear steepening $\sim$ dispersive spreading
 - ▶ This balance produces solitons

Without this balance:

- ▶ Too much nonlinearity $\Rightarrow$ breaking
- ▶ Too much dispersion $\Rightarrow$ spreading

Derivation Strategy (Outline)

1. Solve Laplace equation for ϕ
2. Expand in small parameter δ
3. Express velocity at surface
4. Insert into boundary conditions
5. Expand in ϵ
6. Retain terms up to $O(\epsilon, \delta^2)$

Origin of the $\frac{h^2}{6}$ Dispersive Term

Key idea: dispersion comes from solving Laplace's equation in finite depth.

We solve

$$\phi_{xx} + \phi_{yy} = 0, \quad -h < y < \eta(x, t)$$

with

$$\phi_y = 0 \quad \text{at } y = -h$$

Expand ϕ about the bottom using the boundary condition:

$$\phi(x, y, t) = \Phi(x, t) + \frac{(y+h)^2}{2!} \Phi_{xx} + \frac{(y+h)^4}{4!} \Phi_{xxxx} + \dots$$

Evaluate horizontal velocity at the surface ($y \approx 0$):

$$\phi_x(x, 0, t) = \Phi_x + \frac{h^2}{2} \Phi_{xxx} + \frac{h^4}{24} \Phi_{xxxxx} + \dots$$

From Expansion to Dispersion

Insert expansion into kinematic and dynamic conditions.

After asymptotic reduction (keeping $O(\delta^2)$ terms):

$$\eta_t + c_0 \eta_x + \frac{3c_0}{2h} \eta \eta_x + \frac{c_0 h^2}{6} \eta_{xxx} = 0$$

Where does $\frac{h^2}{6}$ come from?

- ▶ The h^2 originates from vertical structure of ϕ
- ▶ The factor $\frac{1}{6}$ comes from:
 - ▶ combining Taylor coefficients
 - ▶ contributions from both boundary conditions

Physical meaning:

- ▶ Finite depth $\Rightarrow$ vertical variation in velocity
- ▶ This produces wavelength-dependent speed (dispersion)

Resulting KdV Equation (Physical Variables)

The free surface $\eta(x, t)$ satisfies:

$$\eta_t + c_0 \eta_x + \frac{3c_0}{2h} \eta \eta_x + \frac{c_0 h^2}{6} \eta_{xxx} = 0$$

where

$$c_0 = \sqrt{gh}$$

Interpretation:

- ▶ $c_0 \eta_x$: linear wave propagation
- ▶ $\eta \eta_x$: nonlinear steepening
- ▶ η_{xxx} : dispersion from finite depth

Physical Interpretation of Terms

- ▶ **Linear term:**

$$\eta_t + c_0 \eta_x$$

waves travel at shallow water speed $\sqrt{gh}$

- ▶ **Nonlinear term:**

$$\frac{3c_0}{2h} \eta \eta_x$$

larger waves move faster $\Rightarrow$ steepening

- ▶ **Dispersive term:**

$$\frac{c_0 h^2}{6} \eta_{xxx}$$

different wavelengths travel at different speeds

From Physical to Standard KdV

Introduce scaled variables:

$$x' = \frac{x - c_0 t}{\lambda}, \quad t' = \frac{c_0 \delta^2}{\lambda} t, \quad u = \frac{\eta}{a}$$

After rescaling and dropping primes:

$$u_t + 6uu_x + u_{xxx} = 0$$

How to Move Between Variants

General KdV form:

$$u_t + \alpha uu_x + \beta u_{xxx} + \gamma u_x = 0$$

Transformations:

- ▶ Remove linear advection:

$$x \mapsto x - \gamma t$$

- ▶ Normalize nonlinearity:

$$u \mapsto \frac{6}{\alpha} u$$

- ▶ Normalize dispersion:

$$x \mapsto \left(\frac{\beta}{1}\right)^{1/3} x, \quad t \mapsto \left(\frac{\beta}{1}\right) t$$

These reduce any KdV-type equation to:

$$u_t + 6uu_x + u_{xxx} = 0$$

Connection Back to Physics

Undoing scalings:

- ▶ Solution amplitude $\sim a$
- ▶ Width $\sim \lambda$
- ▶ Speed:

$$c = \sqrt{gh} \left(1 + \frac{a}{2h} \right)$$

Key point:

- ▶ Soliton speed increases with amplitude
- ▶ Directly observed by Russell

Limitations of KdV

- ▶ One-dimensional approximation
- ▶ Weak nonlinearity
- ▶ Long wave assumption
- ▶ Flat bottom

To include weak transverse effects $\Rightarrow$ KP equations

KP Equation (Physical Form)

Allow weak dependence in y direction.

$$\left(\eta_t + c_0 \eta_x + \frac{3c_0}{2h} \eta \eta_x + \frac{c_0 h^2}{6} \eta_{xxx} \right)_x + \frac{c_0}{2} \eta_{yy} = 0$$

Standard KP Equations

After scaling:

KP-I:

$$(u_t + 6uu_x + u_{xxx})_x - u_{yy} = 0$$

KP-II:

$$(u_t + 6uu_x + u_{xxx})_x + u_{yy} = 0$$

Physical distinction:

- ▶ KP-I: strong surface tension
- ▶ KP-II: weak surface tension (water waves)

Line Solitons in KP

A **line soliton** is a solution of the form:

$$u(x, y, t) = U(x - ct)$$

Key features:

- ▶ Independent of y
- ▶ Essentially a KdV soliton extended in the transverse direction
- ▶ Forms a planar wave crest

Physical interpretation:

- ▶ Long-crested shallow water wave
- ▶ Nearly uniform in transverse direction

Stability:

- ▶ KP-II: stable
- ▶ KP-I: unstable (transverse breakup)

Lump Solitons in KP

A **lump soliton** is a fully localized solution:

$$u(x, y, t) \rightarrow 0 \quad \text{as } \sqrt{x^2 + y^2} \rightarrow \infty$$

Key features:

- ▶ Localized in both x and y
- ▶ Algebraic decay (not exponential)
- ▶ Exists only for KP-I

Typical structure: $u(x, y, t) \sim \frac{1}{(x-ct)^2 + y^2 + 1}$

Physical interpretation:

- ▶ Localized “blob” of wave energy
- ▶ No 1D analogue in KdV

Key contrast:

- ▶ Line soliton: 1D structure extended in y
- ▶ Lump soliton: intrinsically 2D object

Physical Interpretation of KP

- ▶ Extends KdV to 2D weakly transverse waves
- ▶ Retains:
 - ▶ Nonlinearity
 - ▶ Dispersion
- ▶ Adds:
 - ▶ Transverse spreading (u_{yy})

Key effect:

- ▶ Stability of line solitons depends on KP-I vs KP-II

Summary

- ▶ KdV arises from shallow water with depth h
- ▶ Balance:
$$\text{nonlinearity} \sim \text{dispersion}$$
- ▶ Scaling $\epsilon \sim \delta^2$ is essential
- ▶ Different forms of KdV are related by simple scalings
- ▶ KP extends KdV to weakly 2D waves